

Advanced Data IO

Data Wrangling in R

Google Sheets

gapminder

☆

🔗

☁

File

Edit

View

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Format

Data

Tools

Add-ons

Help

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100%

View only

A1

fx

country

	A	B	C	D	E	F
1	country	continent	year	lifeExp	pop	gdpPercap
2	Australia	Oceania	1952	69.12	8691212	10039.59564
3	Australia	Oceania	1957	70.33	9712569	10949.64959
4	Australia	Oceania	1962	70.93	10794968	12217.22686
5	Australia	Oceania	1967	71.1	11872264	14526.12465
6	Australia	Oceania	1972	71.93	13177000	16788.62948
7	Australia	Oceania	1977	73.49	14074100	18334.19751
8	Australia	Oceania	1982	74.74	15184200	19477.00928
9	Australia	Oceania	1987	76.32	16257249	21888.88903
10	Australia	Oceania	1992	77.56	17481977	23424.76683
11	Australia	Oceania	1997	78.83	18565243	26997.93657
12	Australia	Oceania	2002	80.37	19546792	30687.75473
13	Australia	Oceania	2007	81.235	20434176	34435.36744
14	New Zealand	Oceania	1952	69.39	1994794	10556.57566
15	New Zealand	Oceania	1957	70.26	2229407	12217.39532

☰

Africa

Americas

Asia

Europe

Oceania

Explore

https://docs.google.com/spreadsheets/d/1U6Cf_qEOhiR9AZqTqS3mbMF3zt2db48ZP5v3rkrAEJY/edit#gid=78

Reading data with the googlesheets4 package

First, set up Google credentials.

```
library(googlesheets4)
```

```
# Prompts a browser pop-up  
gs4_auth()
```

When prompted:

- “OK to cache OAuth access credentials in the folder between R sessions?”, select YES
- Make sure you select “allow” on your Google Account for the Tidyverse API.

Reading data with the googlesheets4 package

```
# Once set up, you can automate this process by passing your email  
gs4_auth(email = "avamariehoffman@gmail.com")
```

Reading data with the googlesheets4 package

You can also supply an authorization token directly, but make sure never to make these visible online!

```
library(googledrive)
drive_auth(email= "<email>",
           token = readRDS("google-sheets-token.rds")) # Saved in a file
```

Reading data with the googlesheets4 package

Read in using `read_sheet()`

```
sheet_url <-  
  "https://docs.google.com/spreadsheets/d/1U6Cf_qE0hiR9AZqTqS3mbMF3zt2db48ZP5v3rkrAEJY/edit#gid=780868077"  
sheet_dat_1 <- read_sheet(sheet_url)
```

✓ Reading from "gapminder".

✓ Range 'Africa'.

```
head(sheet_dat_1)
```

```
# A tibble: 6 × 6  
  country continent year lifeExp      pop gdpPercap  
  <chr>    <chr>    <dbl>   <dbl>   <dbl>   <dbl>  
1 Algeria Africa    1952    43.1  9279525    2449.  
2 Algeria Africa    1957    45.7 10270856    3014.  
3 Algeria Africa    1962    48.3 11000948    2551.  
4 Algeria Africa    1967    51.4 12760499    3247.  
5 Algeria Africa    1972    54.5 14760787    4183.  
6 Algeria Africa    1977    58.0 17152804    4910.
```

Reading data with the googlesheets4 package

Specify the sheet name if necessary:

```
sheet_dat_oceania <- read_sheet(sheet_url, sheet = "Oceania")
```

✓ Reading from "gapminder".

✓ Range "'Oceania'".

```
head(sheet_dat_oceania)
```

```
# A tibble: 6 × 6
  country    continent  year lifeExp      pop gdpPercap
  <chr>      <chr>    <dbl>  <dbl>    <dbl>    <dbl>
1 Australia Oceania   1952   69.1  8691212  10040.
2 Australia Oceania   1957   70.3  9712569  10950.
3 Australia Oceania   1962   70.9 10794968  12217.
4 Australia Oceania   1967   71.1 11872264  14526.
5 Australia Oceania   1972   71.9 13177000  16789.
6 Australia Oceania   1977   73.5 14074100  18334.
```


Pull in a subset of data: rows

```
read_sheet(sheet_url, sheet = "Oceania", range = cell_rows(1:4))
```

✓ Reading from "gapminder".

✓ Range "'Oceania'!1:4'.

A tibble: 3 × 6

	country	continent	year	lifeExp	pop	gdpPercap
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Australia	Oceania	1952	69.1	8691212	10040.
2	Australia	Oceania	1957	70.3	9712569	10950.
3	Australia	Oceania	1962	70.9	10794968	12217.

Pull in a subset of data: columns

```
read_sheet(sheet_url, sheet = "Oceania", range = cell_cols("A:B"))
```

✓ Reading from "gapminder".

✓ Range "'Oceania'!A:B".

```
# A tibble: 24 × 2
  country    continent
  <chr>      <chr>
1 Australia Oceania
2 Australia Oceania
3 Australia Oceania
4 Australia Oceania
5 Australia Oceania
6 Australia Oceania
7 Australia Oceania
8 Australia Oceania
9 Australia Oceania
10 Australia Oceania
# i 14 more rows
```

Reading data with the googlesheets4 package

List out the sheet names using `sheet_names()`.

```
sheet_names(sheet_url)
```

```
[1] "Africa"    "Americas" "Asia"      "Europe"    "Oceania"
```

Reading data with the googlesheets4 package

Iterate through the sheet names with `purrr`:

```
gapminder_sheets <- sheet_names(sheet_url)
data_list <- map(gapminder_sheets, ~ read_sheet(sheet_url, sheet = .x))
```

- ✓ Reading from "gapminder".
- ✓ Range "'Africa'".
- ✓ Reading from "gapminder".
- ✓ Range "'Americas'".
- ✓ Reading from "gapminder".
- ✓ Range "'Asia'".
- ✓ Reading from "gapminder".
- ✓ Range "'Europe'".
- ✓ Reading from "gapminder".
- ✓ Range "'Oceania'".

Reading data with the googlesheets4 package

Check out the list:

```
str(data_list)
```

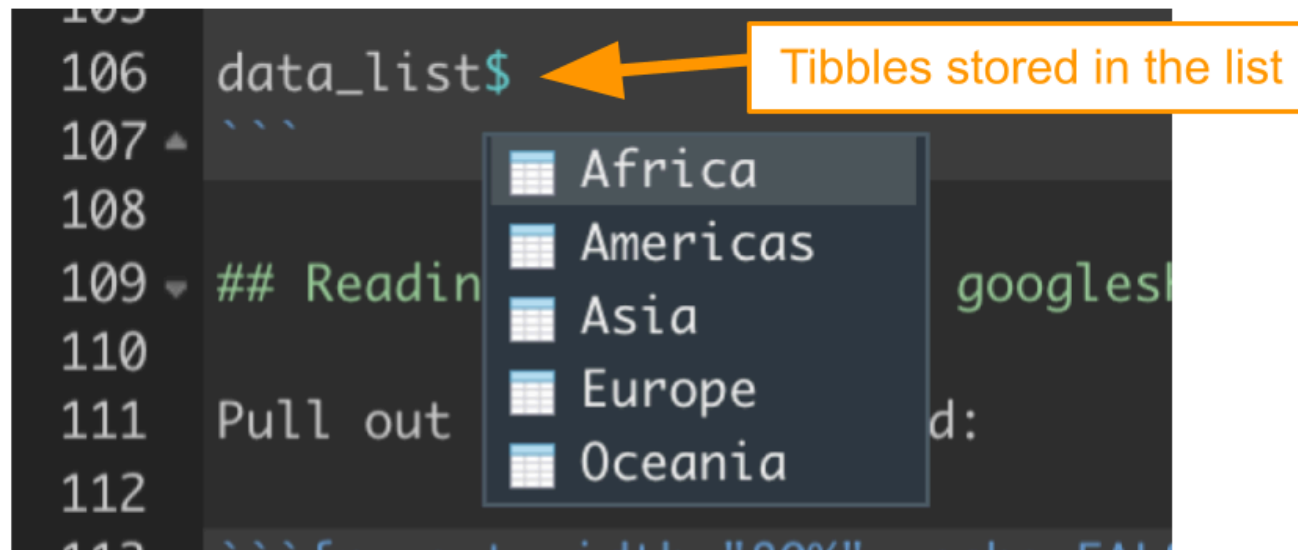
List of 5

```
$ : tibble [624 × 6] (S3: tbl_df/tbl/data.frame)
  ..$ country : chr [1:624] "Algeria" "Algeria" "Algeria" "Algeria" ...
  ..$ continent: chr [1:624] "Africa" "Africa" "Africa" "Africa" ...
  ..$ year      : num [1:624] 1952 1957 1962 1967 1972 ...
  ..$ lifeExp   : num [1:624] 43.1 45.7 48.3 51.4 54.5 ...
  ..$ pop       : num [1:624] 9279525 10270856 11000948 12760499 14760787 ...
  ..$ gdpPercap: num [1:624] 2449 3014 2551 3247 4183 ...
$ : tibble [300 × 6] (S3: tbl_df/tbl/data.frame)
  ..$ country : chr [1:300] "Argentina" "Argentina" "Argentina" "Argentina" ...
  ..$ continent: chr [1:300] "Americas" "Americas" "Americas" "Americas" ...
  ..$ year      : num [1:300] 1952 1957 1962 1967 1972 ...
  ..$ lifeExp   : num [1:300] 62.5 64.4 65.1 65.6 67.1 ...
  ..$ pop       : num [1:300] 17876956 19610538 21283783 22934225 24779799 ...
  ..$ gdpPercap: num [1:300] 5911 6857 7133 8053 9443 ...
$ : tibble [396 × 6] (S3: tbl_df/tbl/data.frame)
  ..$ country : chr [1:396] "Afghanistan" "Afghanistan" "Afghanistan" "Afghan" ...
  ..$ continent: chr [1:396] "Asia" "Asia" "Asia" "Asia" ...
  ..$ year      : num [1:396] 1952 1957 1962 1967 1972 ...
  ..$ lifeExp   : num [1:396] 28.8 30.3 32 34 36.1 ...
  ..$ pop       : num [1:396] 8425333 9240934 10267083 11537966 13079460 ...
  ..$ gdpPercap: num [1:396] 779 821 853 836 740 ...
$ : tibble [360 × 6] (S3: tbl_df/tbl/data.frame)
  ..$ country : chr [1:360] "Albania" "Albania" "Albania" "Albania" ...
```

Reading data with the googlesheets4 package

Pull out sheets as needed:

```
data_list[[{sheet}]]  
# OR  
data_list %>% pluck({sheet})  
# OR (if named)  
data_list${sheet}
```



Writing data with the googlesheets4 package

```
sheet_dat_oceania <- data_list[[5]]  
  
sheet_dat_oceania <- sheet_dat_oceania %>%  
  mutate(lifeExp_days = lifeExp * 365)  
  
sheet_out <- gs4_create("Oceania-days",  
                        sheets = list(oceania_days = sheet_dat_oceania))
```

✓ Creating new Sheet: "Oceania-days".

```
# Opens a browser window  
gs4_browse(sheet_out)
```

Append data with the googlesheets4 package

```
sheet_append(sheet_out, data = sheet_dat_oceania, sheet = "Oceania_days")
```

- ✓ Writing to "Oceania-days".
- ✓ Appending 24 rows to 'Oceania_days'.

JHU Tidyverse Book

<https://jhudatascience.org/tidyversecourse/get-data.html#google-sheets>

Lab

<http://sisbid.github.io/Data-Wrangling/labs/advanced-io-lab.Rmd>

JSON: JavaScript Object Notation

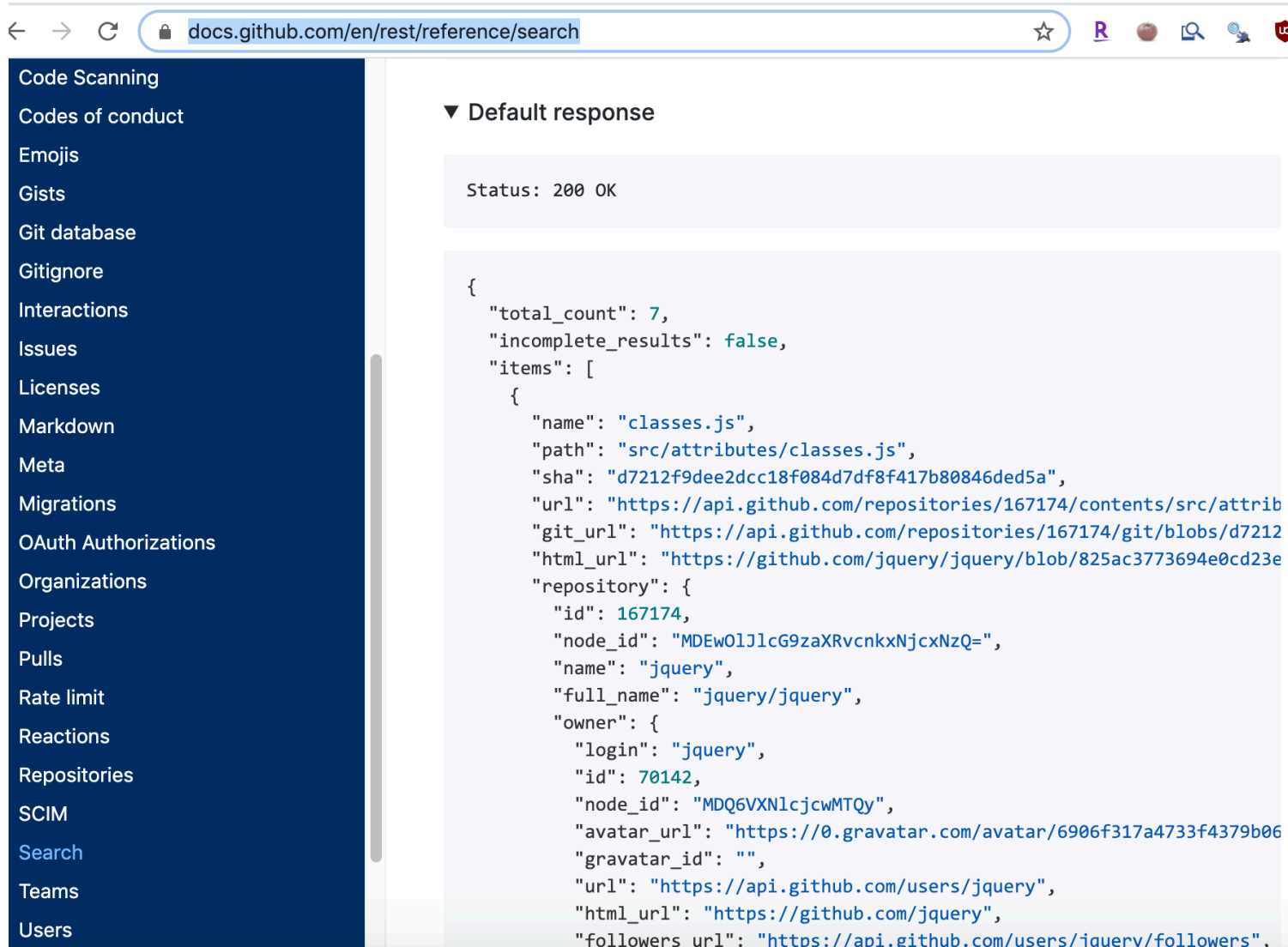
Lists of stuff

Example [\[edit \]](#)

The following example shows a possible JSON representation describing a person.

```
{
  "firstName": "John",
  "lastName": "Smith",
  "isAlive": true,
  "age": 27,
  "address": {
    "streetAddress": "21 2nd Street",
    "city": "New York",
    "state": "NY",
    "postalCode": "10021-3100"
  },
  "phoneNumbers": [
    {
      "type": "home",
      "number": "212 555-1234"
    },
    {
      "type": "office",
      "number": "646 555-4567"
    }
  ],
  "children": [],
  "spouse": null
}
```

Why JSON matters



The screenshot shows the GitHub REST API reference page for the search endpoint. The left sidebar contains a navigation menu with links to various GitHub features. The main content area displays the 'Default response' for the search endpoint, which is a JSON object. The status is '200 OK'. The JSON response includes a 'total_count' of 7, 'incomplete_results' set to false, and an 'items' array containing one object for the repository 'jquery/jquery'.

Code Scanning
Codes of conduct
Emojis
Gists
Git database
Gitignore
Interactions
Issues
Licenses
Markdown
Meta
Migrations
OAuth Authorizations
Organizations
Projects
Pulls
Rate limit
Reactions
Repositories
SCIM
Search
Teams
Users

▼ Default response

Status: 200 OK

```
{
  "total_count": 7,
  "incomplete_results": false,
  "items": [
    {
      "name": "classes.js",
      "path": "src/attributes/classes.js",
      "sha": "d7212f9dee2dcc18f084d7df8f417b80846ded5a",
      "url": "https://api.github.com/repositories/167174/contents/src/attrib",
      "git_url": "https://api.github.com/repositories/167174/git/blobs/d7212",
      "html_url": "https://github.com/jquery/jquery/blob/825ac3773694e0cd23e",
      "repository": {
        "id": 167174,
        "node_id": "MDEwO1JlcG9zaXRvcnkxNjcxNzQ=",
        "name": "jquery",
        "full_name": "jquery/jquery",
        "owner": {
          "login": "jquery",
          "id": 70142,
          "node_id": "MDQ6VXNlcjcwMTQy",
          "avatar_url": "https://0.gravatar.com/avatar/6906f317a4733f4379b06",
          "gravatar_id": "",
          "url": "https://api.github.com/users/jquery",
          "html_url": "https://github.com/jquery",
          "followers_url": "https://api.github.com/users/jquery/followers".
        }
      }
    }
  ]
}
```

<https://docs.github.com/en/rest/reference/search>

```
#install.packages("jsonlite")
library(jsonlite)
jsonData <-
  fromJSON("https://raw.githubusercontent.com/Biuni/PokemonGO-Pokedex/master/pokedex.json")
head(jsonData)
```

```
$pokemon
      id num      name      img
1      1 001  Bulbasaur http://www.serebii.net/pokemongo/pokemon/001.png
2      2 002   Ivysaur http://www.serebii.net/pokemongo/pokemon/002.png
3      3 003  Venusaur http://www.serebii.net/pokemongo/pokemon/003.png
4      4 004 Charmander http://www.serebii.net/pokemongo/pokemon/004.png
5      5 005 Charmeleon http://www.serebii.net/pokemongo/pokemon/005.png
6      6 006  Charizard http://www.serebii.net/pokemongo/pokemon/006.png
7      7 007   Squirtle http://www.serebii.net/pokemongo/pokemon/007.png
8      8 008  Wartortle http://www.serebii.net/pokemongo/pokemon/008.png
9      9 009  Blastoise http://www.serebii.net/pokemongo/pokemon/009.png
10     10 010  Caterpie http://www.serebii.net/pokemongo/pokemon/010.png
11     11 011  Metapod http://www.serebii.net/pokemongo/pokemon/011.png
12     12 012 Butterfree http://www.serebii.net/pokemongo/pokemon/012.png
13     13 013   Weedle http://www.serebii.net/pokemongo/pokemon/013.png
14     14 014   Kakuna http://www.serebii.net/pokemongo/pokemon/014.png
15     15 015  Beedrill http://www.serebii.net/pokemongo/pokemon/015.png
16     16 016   Pidgey http://www.serebii.net/pokemongo/pokemon/016.png
17     17 017 Pidgeotto http://www.serebii.net/pokemongo/pokemon/017.png
18     18 018   Pidgeot http://www.serebii.net/pokemongo/pokemon/018.png
19     19 019  Rattata http://www.serebii.net/pokemongo/pokemon/019.png
20     20 020  Raticate http://www.serebii.net/pokemongo/pokemon/020.png
21     21 021  Spearow http://www.serebii.net/pokemongo/pokemon/021.png
22     22 022   Fearow http://www.serebii.net/pokemongo/pokemon/022.png
23     23 023    Ekans http://www.serebii.net/pokemongo/pokemon/023.png
```

Data frame structure from JSON

```
dim(jsonData$pokemon)
```

```
[1] 151 17
```

```
class(jsonData$pokemon)
```

```
[1] "data.frame"
```

```
jsonData$pokemon %>% filter(type == "Fire") %>% select(!(img))
```

	id	num	name	type	height	weight		candy	candy_count
1	4	004	Charmander	Fire	0.61 m	8.5 kg	Charmander	Candy	25
2	5	005	Charmeleon	Fire	1.09 m	19.0 kg	Charmander	Candy	100
3	37	037	Vulpix	Fire	0.61 m	9.9 kg	Vulpix	Candy	50
4	38	038	Ninetales	Fire	1.09 m	19.9 kg	Vulpix	Candy	NA
5	58	058	Growlithe	Fire	0.71 m	19.0 kg	Growlithe	Candy	50
6	59	059	Arcanine	Fire	1.91 m	155.0 kg	Growlithe	Candy	NA
7	77	077	Ponyta	Fire	0.99 m	30.0 kg	Ponyta	Candy	50
8	78	078	Rapidash	Fire	1.70 m	95.0 kg	Ponyta	Candy	NA
9	126	126	Magmar	Fire	1.30 m	44.5 kg		None	NA
10	136	136	Flareon	Fire	0.89 m	25.0 kg	Eevee	Candy	NA

		egg	spawn_chance	avg_spawns	spawn_time	multipliers
1		2 km	0.2530	25.30	08:45	1.65
2	Not in	Eggs	0.0120	1.20	19:00	1.79
3		5 km	0.2200	22.00	13:43	2.74, 2.81
4	Not in	Eggs	0.0077	0.77	01:32	NULL
5		5 km	0.9200	92.00	03:57	2.31, 2.36
6	Not in	Eggs	0.0170	1.70	03:11	NULL

Going deeper..

```
class(jsonData$pokemon$type) # Can be lists
```

```
[1] "list"
```

```
jsonData$pokemon$type
```

```
[[1]]  
[1] "Grass"  "Poison"
```

```
[[2]]  
[1] "Grass"  "Poison"
```

```
[[3]]  
[1] "Grass"  "Poison"
```

```
[[4]]  
[1] "Fire"
```

```
[[5]]  
[1] "Fire"
```

```
[[6]]  
[1] "Fire"  "Flying"
```

```
[[7]]  
[1] "Water"
```

```
[[8]]
```


Data frame structure from JSON

```
class(jsonData$pokemon$next_evolution[[1]]) # Or lists of data.frames!
```

```
[1] "data.frame"
```

```
jsonData$pokemon$next_evolution
```

```
[[1]]  
  num      name  
1 002 Ivysaur  
2 003 Venusaur
```

```
[[2]]  
  num      name  
1 003 Venusaur
```

```
[[3]]  
NULL
```

```
[[4]]  
  num      name  
1 005 Charmeleon  
2 006 Charizard
```

```
[[5]]  
  num      name  
1 006 Charizard
```

```
[[6]]  
NULL
```

Summary

- `googlesheets4`: interface with Google Sheets
 - use `gs4_auth()` to stash and set up credentials
 - use `read_sheet()` and other functions to bring data in
- `jsonlite`: make sense of .json data
 - use `$` notation to explore structure

<http://sisbid.github.io/Data-Wrangling/labs/advanced-io-lab.Rmd>

Extra Slides: Web Scraping and APIs

View the source

Please note that to use the ExpressionSets below, you will need to install [Bioconductor](#) and run the command `library(Biobase)`

✧ The Datasets

Study	PMID	Species	Number of biological replicates	Number of uniquely aligned reads	Exp				otype	Notes	
bodymap	not published, but publicly available here	human	19	2,197,622,796	link					Illumina Human BodyMap 2.0 - tissue comparison	
cheung	20856902	human	41	834,584,950	link					HapMap - CEU	
core	19056941	human	2	8,670,342	link					lung fibroblasts	
gilad	20009012	human	6	41,356,738	link					liver; males and femlaes	
maqc	20167110	human	14 (technical)** 2 (biological)	71,970,164	original pooled	original pooled	original pooled				experiment: MAQC-2
montgomery	20220756	human	60	*886,468,054	link	link	link				HapMap - CEU
pickrell	20220758	human	69	*886,468,054	link	link	link				HapMap - YRI
sultan	18599741	human	4	6,573,643	link	link	link				cell type comparison
wang	18978772	human	22	223,929,919	link	link	link				tissue comparison
katz.mouse	21057496	mouse	4	14.368.471	link	link	link				control vs. CUG-BP1

Back

Forward

Reload

Save As...

Print...

Cast...

Translate to English

 Block element...

View Page Source

View Frame Source

Reload Frame

Inspect

Speech Services

What the computer sees

```
1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
2 <html xmlns="http://www.w3.org/1999/xhtml">
3 <head>
4 <script src="sorttable.js" type="text/javascript"></script>
5 <title>ReCount: analysis-ready RNA-seq gene count datasets</title>
6 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
7 <link rel="stylesheet" type="text/css" href="css/style.css" media="screen" />
8 <script type="text/javascript">
9
10     var _gaq = _gaq || [];
11     _gaq.push(['_setAccount', 'UA-26478269-2']);
12     _gaq.push(['_trackPageview']);
13
14     (function() {
15         var ga = document.createElement('script'); ga.type = 'text/javascript'; ga.async = true;
16         ga.src = ('https:' == document.location.protocol ? 'https://ssl' : 'http://www') + '.google-analytics.com/ga.js';
17         var s = document.getElementsByTagName('script')[0]; s.parentNode.insertBefore(ga, s);
18     })();
19
20 </script>
21
22 </head>
23
24 <body class="c20">
25 <div id="wrap">
26     <div id="top">
27         <div class="lefts">
28             <table width="100%" cellpadding="2">
29                 <tr><td>
30                     <a href="./index.shtml"><h1>ReCount</h1></a>
31                     <h2>A multi-experiment resource of analysis-ready RNA-seq gene count datasets</h2>
32                 </td><td align="right" valign="middle">
33                     <h1><a href="http://www.biostat.jhsph.edu/"></img></a>&nbsp;&nbsp;&nbsp;</h1>
34                 </td></tr>
35             </table>
36         </div>
37     </div>
38
39     <div id="subheader">
40 <p><b>There is now <a href="https://jhubiostatistics.shinyapps.io/recount/">a new version of recount</a> that provides processed and summarized express
data for nearly 60,000 human RNA-seq samples from the Sequence Read Archive (SRA). The <a href="https://github.com/leekgroup/recount">associated
Bioconductor package</a> provides a convenient API for querying, downloading, and analyzing the data. Each processed study consists of meta- and phenot
data, the expression levels of genes and their underlying exons and splice junctions, and corresponding genomic annotation. See <a
```

Ways to see the source

Chrome:

1. right click on page
2. select "View Page Source"

Firefox:

1. right click on page
2. select "View Page Source"

Microsoft Edge:

1. right click on page
2. select "view source"

Safari

1. click on "Safari"
2. select "Preferences"
3. go to "Advanced"
4. check "Show Develop menu in menu bar"
5. right click on page
6. select "View Page Source"

<https://github.com/simonmunzert/rscraping-jsm-2016/blob/c04fd91fec711df65c838e07723125155a7f2cda/02-scraping-with-rvest.r>

Inspect element

← → ↻ ⓘ Not Secure | bowtie-bio.sourceforge.net/recount/ ☆ R 🔍 🔍 🔒 📄 📄 W. 📺 📺 🔍 🔍 ⚙️ ⚙️ 📄 📄 📄 📄

Please note that to use the ExpressionSets below, you will need to install [Bioconductor](#) and run the command `library(Biobase)`.

❖ The Datasets

Study	PMID	Species	Number of biological replicates	Number of uniquely aligned reads	Expression		Notes
maq	20167110	human	14 (technical)** 2 (biological)	71,970,164	original pooled		experiment: MAQC-2
modencodefly	21179090	fly	147 (technical)** 30 (biological)	2,278,788,557	original pooled		developmental time course
modencodeworm	19181841	worm	46	1,451,119,823	link		developmental time course
hammer	20452967	rat	8	158,178,477	link	link link	experimental vs. control at 2 time points
nagalakshmi	18451266	yeast	4	7,688,602	link	link link	priming technique comparison
bottomly	21455293	mouse	21	343,445,340	link	link link	2 inbred mouse strains
yang	20363980	mouse	1	27,883,862	link	link link	hybrid cell line, X always inactive
trapnell	20436464	mouse	4	111,376,152	link	link link	time course
mortazavi	18516045	mouse	3	61,732,881	link	link link	tissue comparison

Copy XPath

← → ↻ ⓘ Not Secure | bowtie-bio.sourceforge.net/recount/ ☆ R 🔍 🔍 🔒 📄 🔄 W. ▶ 🔍 ⚙️ 📱 📺 📶

The Datasets 888.75 × 1106.8

Study	PMID	Species	Number of biological replicates	Number of uniquely aligned reads	ExpressionSet	Count table	Phenotype table	Notes
bodymap	not published, but publicly available here	human	19	2,197,622,796	link	link	link	Illumina Human BodyMap 2.0 - tissue comparison
cheung	20856902	human	41	834,584,950	link	link	link	HapMap - CEU
core	19056941	human	2	8,670,342	link	link	link	lung fibroblasts
gilad	2000			41,356,738	link	link	link	liver; males and females

Elements Console Sources Application Security Lighthouse

Brief description of experiment

Please note that to use the t
<a href="http://www.biocondu
" and run the command "
<tt>library(Biobase)</tt>
<h3>The Datasets</h3>
.. <div id="recounttab"> == \$0
 <table class="sortable">
 <thead>
 <tr>

html body.c20.vsc-initialized div#wrap div#m

Console What's New ✕

Highlights from the Chrome 83 update

Update your definitions from the DevTools

Styles Computed Event Listeners DOM Breakpoints >>

Filter :hov .cls +

element.style {
}

* {
padding: > 0;
margin: > 0;
} style.css:7

div {
display: block;
} user agent stylesheet

Inherited from div#leftside

#leftside {
padding-left: 8px;
color: #555;
} style.css:143

Copy XPath

Use SelectorGadget

<https://rvest.tidyverse.org/articles/selectorgadget.html>

rvest package

```
recount_url <- "http://bowtie-bio.sourceforge.net/recount/"
# install.packages("rvest")
library(rvest)
htmlfile <- read_html(recount_url)

nds <- html_nodes(htmlfile, xpath = '//*[@id="recounttab"]/table')
dat <- html_table(nds)
dat <- as.data.frame(dat)
head(dat)
```

	X1		X2	X3
1	Study		PMID	Species
2	bodymap	not published, but publicly available here		human
3	cheung		20856902	human
4	core		19056941	human
5	gilad		20009012	human
6	maq		20167110	human

	X4	X5
1	Number of biological replicates	Number of uniquely aligned reads
2	19	2,197,622,796
3	41	834,584,950
4	2	8,670,342
5	6	41,356,738
6	14 (technical)** 2 (biological)	71,970,164

	X6	X7	X8
1	ExpressionSet	Count table	Phenotype table
2	link	link	link
3	link	link	link
4	link	link	link

Little cleanup

```
colnames(dat) <- as.character(dat[1,])
dat <- dat[-1,]
head(dat)
```

	Study	PMID	Species
2	bodymap not published, but publicly available here		human
3	cheung	20856902	human
4	core	19056941	human
5	gilad	20009012	human
6	maq	20167110	human
7	montgomery	20220756	human
	Number of biological replicates	Number of uniquely aligned reads	
2	19	2,197,622,796	
3	41	834,584,950	
4	2	8,670,342	
5	6	41,356,738	
6	14 (technical)** 2 (biological)	71,970,164	
7	60	*886,468,054	
	ExpressionSet	Count table	Phenotype table
2	link	link	link
3	link	link	link
4	link	link	link
5	link	link	link
6	original pooled original pooled original pooled		
7	link	link	link
	Notes		
2	Illumina Human BodyMap 2.0 -- tissue comparison		
3	HapMap - CEU		
4	lung fibroblasts		

Ethics and Web Scraping

<https://slate.com/culture/2020/04/whitney-museum-new-york-apartment-exhibit-creators-interview.html>

BROW BEAT

An Art Exhibit You Can Visit Without Leaving Your Couch

The creators of the Whitney Museum's *New York Apartment* explain how they combined thousands of listings into a website for one massive, \$43.9 billion dwelling.

BY RACHELLE HAMPTON

APRIL 02, 2020 • 7:30 AM

NEW YORK APARTMENT

FOR SALE

\$43,869,676,331

65,764 bedrooms

55,588 bathrooms

36,672,535 sq ft

Are you constantly frustrated with seeing the same low-grade renovations and design choices in townhouses?

Are you in love with you-we?

Are you looking for a charming well-priced 1 Bedroom home in a sought-after area?

Are you looking for a very, elegant home closer to all that is held dear to a New Yorker?

Are you looking for a deal?

Are you looking for a great investment home?

Are you looking for a full-like open space with both light, view, and luxury condo services?

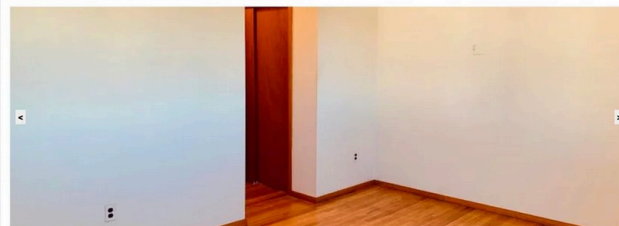
Are you looking for Beautiful Open Views?

Are you looking for that House that is in your Price Range, is DETACHED, has 3 Bedrooms, 4 Bathrooms, One Garage, & a Yard all in the Prime Bay Section of Staten Island with Water View & NOT in a FLOOD ZONE surrounded by Million Dollar Homes?

Are you looking for that perfect balance of convenience and security?

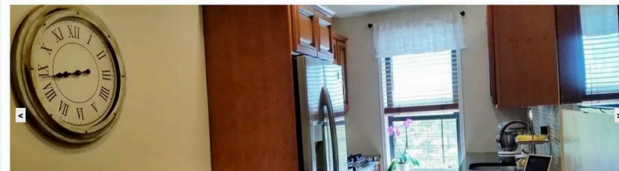
Are you looking for the perfect primary residence, find a home or investment property?

Are you looking to do a Lease or Rent to Own Home?



Bedrooms

1 of 20753



Ethics and Web Scraping

<https://doi.org/10.1016/j.dib.2020.106178>



Contents lists available at [ScienceDirect](#)

Data in Brief

journal homepage: www.elsevier.com/locate/dib



Data Article

COVID-19: A scholarly production dataset report for research analysis



Breno Santana Santos^{a,b,*}, Ivanovitch Silva^a,
Marcel da Câmara Ribeiro-Dantas^c, Gisliany Alves^a,
Patricia Takako Endo^d, Luciana Lima^a

^a Universidade Federal do Rio Grande do Norte (UFRN), Rio Grande do Norte, Brazil

^b Núcleo de Pesquisa e Prática em Inteligência Competitiva (NUPIC), Universidade Federal de Sergipe (UFS), Itabaiana, SE, Brazil

^c Institut Curie (UMR168), Sorbonne Université (EDITE), Paris, France

^d Universidade de Pernambuco (UPE), Pernambuco, Brazil

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Scientometrics

ABSTRACT

COVID-2019 has been recognized as a global threat, and several studies are being conducted in order to contribute to the fight and prevention of this pandemic. This work presents a scholarly production dataset focused on COVID-19, providing an overview of scientific research activities, making it possible to identify countries, scientists and research groups most active in this task force to combat the coronavirus disease. The dataset is composed of 40,212 records of articles' metadata collected from Scopus, PubMed, arXiv and bioRxiv databases from January 2019 to July 2020. Those data were extracted by using the techniques of Python Web Scraping and preprocessed with Pandas Data Wrangling. In addition,

Ethics and Web Scraping

<https://techcrunch.com/2017/04/28/someone-scraped-40000-tinder-selfies-to-make-a-facial-dataset-for-ai-experiments/>



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Someone scraped 40,000 Tinder selfies to make a facial dataset for AI experiments



Natasha Lomas @riptari / 7:21 PM EDT • April 28, 2017

 Comment

Tinder users have many motives for uploading their likeness to the dating app. But contributing a facial biometric to a downloadable data set for training convolutional neural networks probably wasn't top of their list when they signed up to swipe.

A user of Kaggle, a platform for machine learning and data science competitions which was [recently acquired by Google](#), has uploaded a facial data set he says was created by exploiting Tinder's API to scrape 40,000 profile photos from Bay Area users of the dating app — 20,000 apiece from profiles of each gender.

The data set, called [People of Tinder](#), consists of six downloadable zip files, with four containing around 10,000 profile photos each and two files with sample sets of around 500 images per gender.

Some users have had multiple photos scraped from their profiles, so there is likely a lot fewer

Ethics and Web Scraping

<https://on.wsj.com/3hzeu9i>

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Alibaba Falls Victim to Chinese Web Crawler in Large Data Leak

Software developer scrapes 1.1 billion pieces of user data, including IDs and phone numbers, over eight months



In less than six months, China's tech giant Ant went from planning a blockbuster IPO to restructuring in response to pressure from the central bank. As the U.S. also takes aim at big tech, here's how China is moving faster. Photo illustration: Sharon Shi

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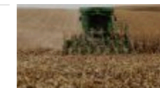
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2. Richard Branson's Virgin Galactic Flight Opens Door to Space Tourism



3. Biden Order Takes Aim at Tractor Repair



4. Why Aren't Millions of Unemployed Americans Finding Jobs?



APIs

Application Programming Interfaces

<https://developers.facebook.com/>

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Take a closer look at the products we offer.

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AR/VR

In biology too!

<http://www.ncbi.nlm.nih.gov/books/NBK25501/>
<https://www.ncbi.nlm.nih.gov/home/develop/api/>



Entrez Programming Utilities Help

Bethesda (MD): [National Center for Biotechnology Information \(US\)](#); 2010-.

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Introduction to the E-utilities

-  [E-utilities Introduction](#)
- Please see the [Release Notes](#) for details and changes.

The Entrez Programming Utilities (E-utilities) are a set of eight server-side programs that provide a stable interface into the Entrez query and database system at the National Center for Biotechnology Information (NCBI). The E-utilities use a fixed URL syntax that translates a standard set of input parameters into the values necessary for various NCBI software components to search for and retrieve the requested data. The E-utilities are therefore the structured interface to the Entrez system, which currently includes 38 databases covering a variety of biomedical data, including nucleotide and protein sequences, gene records, three-dimensional molecular structures, and the biomedical literature.

Contents

☐ [E-utilities Quick Start](#)

Created: December 12, 2008; Last Update: October 24, 2018.

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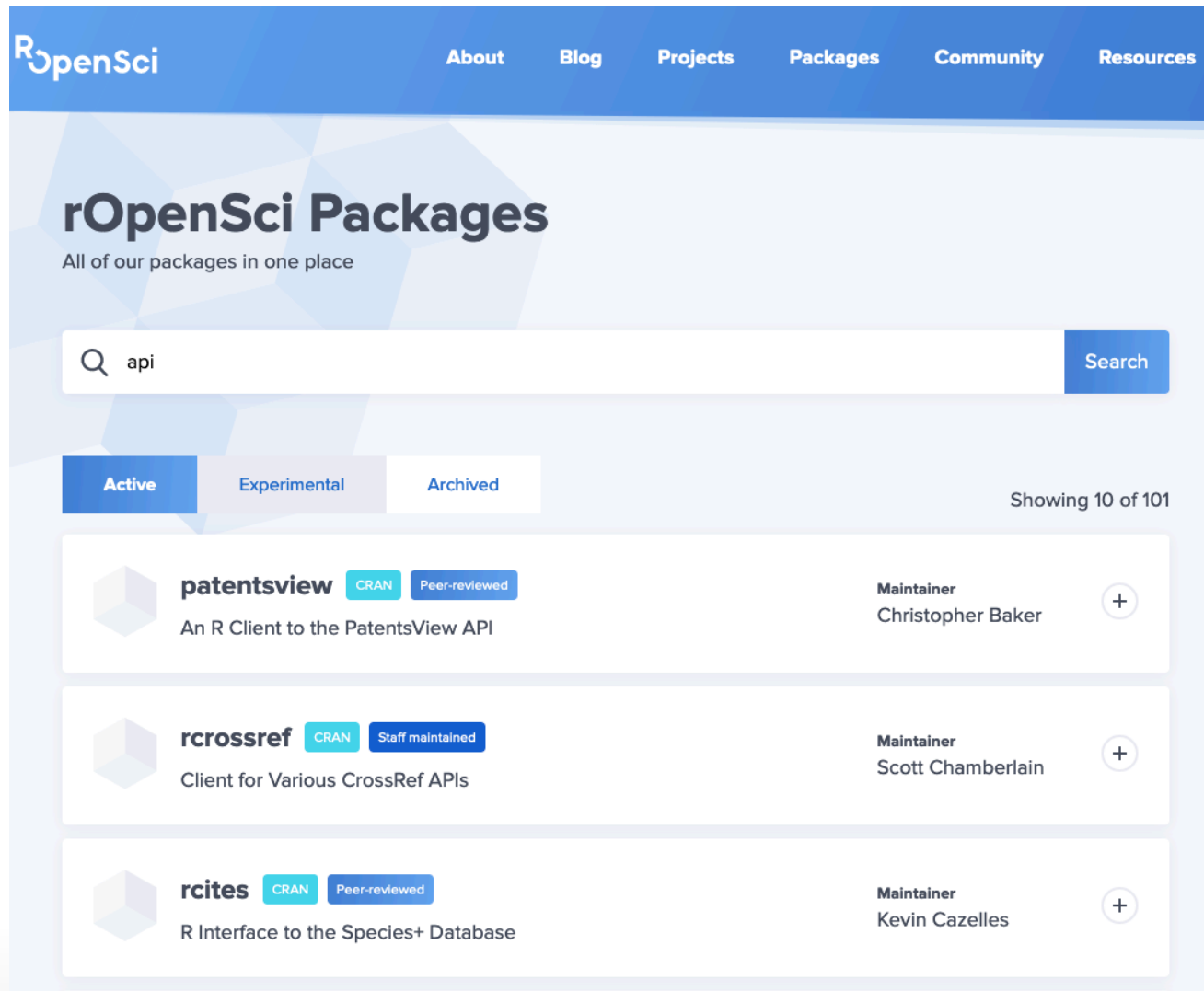
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Step 0: Did someone do this already

<https://ropensci.org/packages/>



The screenshot shows the rOpenSci Packages website. The header is blue with the rOpenSci logo and navigation links: About, Blog, Projects, Packages, Community, and Resources. The main heading is "rOpenSci Packages" with the subtitle "All of our packages in one place". A search bar contains the text "api" and a "Search" button. Below the search bar are three tabs: "Active" (selected), "Experimental", and "Archived". To the right of the tabs, it says "Showing 10 of 101". The list of packages includes:

Package Name	CRAN	Peer-reviewed	Maintainer	Action
patentsview	Yes	Yes	Christopher Baker	+
rcrossref	Yes	Staff maintained	Scott Chamberlain	+
rcites	Yes	Yes	Kevin Cazelles	+

Step 0: Did someone do this already

`tidycensus` package: <https://walker-data.com/tidycensus/articles/basic-usage.html>

tidycensus 1.0.0.9000

[Home](#) [Reference](#) [Articles](#) [Changelog](#)

tidycensus

tidycensus is an R package that allows users to interface with the US Census Bureau's decennial Census and five-year American Community APIs and return tidyverse-ready data frames, optionally with simple feature geometry included. Install from CRAN with the following command:

```
install.packages("tidycensus")
```

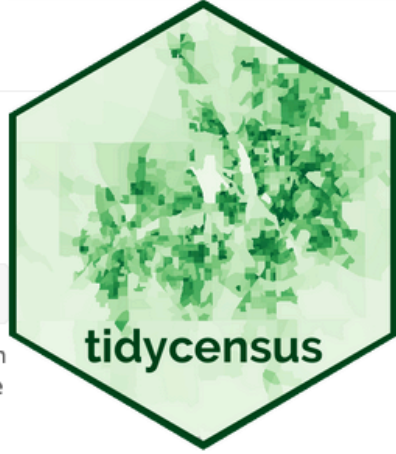
tidycensus is designed to help R users get Census data that is pre-prepared for exploration within the **tidyverse**, and optionally spatially with **sf**. To learn more about how the package works, please read through the following articles:

- [Basic usage of tidycensus](#)
- [Spatial data in tidycensus](#)
- [Margins of error in the ACS](#)
- [Other Census Bureau datasets](#)
- [Working with Census microdata](#)

Future development

To keep up with on-going development of **tidycensus** and get even more examples of how to use the package, subscribe to the Walker Data email list below. You'll also get updates on the forthcoming CRC Press book *Analyzing the US Census with R*, which will cover a wide range of Census data analysis applications.

While **tidycensus** focuses on a select number of US Census Bureau datasets, there are many others available via the Census Bureau API. For access to all of these APIs, please check out Hannah Recht's excellent [censusapi package](#).



Links

Download from CRAN at <https://cloud.r-project.org/package=tidycensus>

Browse source code at <https://github.com/walkerke/tidycensus/>

Report a bug at <https://github.com/walkerke/tidycensus/issues>

License

MIT + file [LICENSE](#)

Developers

Kyle Walker
Author, maintainer

Matt Herman
Author

[All authors...](#)

Dev status

 R-CMD-check passing

Step 1: DIY

<https://github.com/ThatCopy/catAPI/wiki/Usage>

```
#install.packages("httr")  
library(httr)  
  
# Requests a random cat fact  
query_url <- "https://catfact.ninja/fact"  
  
req <- GET(query_url)  
content(req)
```

```
$fact  
[1] "Milk can give some cats diarrhea."
```

```
$length  
[1] 33
```

Not all APIs are “open”

<https://walker-data.com/tidycensus/articles/basic-usage.html>

```
# install.packages("tidycensus")
library(tidycensus)
# Supplied by census.gov
census_api_key("YOUR API KEY GOES HERE")

get_decennial(geography = "state",
               variables = "P013001", # code for median age
               year = 2010)
```